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Protective Effects of *Rasayanas* on Cyclophosphamide- and Radiation-Induced Damage

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ABSTRACT

Rasayanas are a group of herbal drug preparations widely used in Ayurveda to improve the general health of the body. In mice, *Rasayanas* are potent myeloprotective agents against chemotherapeutic agents and radiation. *Rasayanas* are also effective myeloprotectors in patients with cancer undergoing chemotherapy and/or radiotherapy. In this study, we provide further evidence to support the chemoprotective and radioprotective efficacy of four *Rasayanas* in mice. *Rasayanas* were found to reduce the loss of body weight and organ weight induced by cyclophosphamide and radiation significantly. *Rasayanas* were also found to protect tissue from cytotoxic injury associated with reduced serum and liver lipid peroxides, alkaline phosphatase, and glutamate pyruvate transaminase in cyclophosphamide- and radiation-treated animals. These results suggest the potential chemoprotective and radioprotective effects of *Rasayanas,* which require further study to explore their complete usefulness in cancer therapy.

INTRODUCTION

Radiation- and chemical-induced cytoreduction are used therapeutically for the treatment of various neoplasms. However, the efficient use of these treatment strategies has been hindered by untoward reactions and toxicities at effective doses because of their inability to differentiate between normal and malignant cells. Treatment with these agents produces myelosuppression, immunosuppression, and other toxic effects such as nephrotoxicity, neurotoxicity, etc. on normal cells (Halnan, 1982; Yarmonenko, 1988). This may result in impairment of patient's quality of life and reduced tumor control because of the inability to deliver adequate dose-intensive therapy against the cancer. In addition, treatment-induced neutropenia and immunosuppression make these patients more susceptible to infection. Therefore, the use of adjuvants along with chemotherapeutic drugs or radiation was proposed with the goal of reducing toxic effects without altering the outcome of the treatment and improving the efficiency of that particular treatment modality (Hersh, 1982).

Several compounds were identified as having protective effects against chemotherapeutic drugs and radiation-induced toxicity (Links and Lewis, 1999; Vasin, 1999; Weiss, 1997; Weiss and Landauer, 2000). Most of the radioprotectors are sulfhydryl compounds (Yuhas, 1977; Zimmermann and Kimmig, 1998). Several antioxidants such as vitamin E, α-tocopherol, Vitamin C, β-carotene, butylated hydroxytoluene (BHT), ellagic acid, bixine, and

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plant products such as curcumin were also reported to have radioprotective effects (Salvadori et al., 1996; Thresiamma et al., 1996). Protective effects against cytotoxic therapy have also been reported to have been conferred by immunomodulatory substances. It has been suggested that radioprotective and chemoprotective activity conferred by immunomodulators may be attributed to their capacity to enhance hemopoietic and immune functions (Ainsworth, 1988; Hersh, 1982).

Interest in herbal products for the treatment and prevention of cancer has gained momentum in recent years. This is because herbs are effective when given orally, have low preparation cost, are nontoxic, and can be administered for long periods of time. Studies have shown that several immunomodulatory and antioxidant plant products can reduce the toxic effects of radiation and chemotherapy. These include curcumin/turmeric (Thressiamma et al., 1996), garlic (Ruby et al., 1995; Unnikrishnan et al., 1990) *Withania somnifera* (Davis and Kuttan, 1999, 2000, *Vitis vinifera* (Castillo et al., 2000), β-carotene (Salvadori et al., 1996), *Lycium chinense* (Hsu et al., 1999), *Viscum album* (Kuttan and Kuttan, 1993) *Ocimum flavanoids* (Prakash and Gupta, 2000; Uma Devi et al., 2000), etc.

Rasayanas are a group of complex herbal drug preparations widely used in Ayurveda, an Indian traditional system of medicine, to improve the general health of the body (for review, see Vayalil et al., 2002). It has been shown that *Rasayanas* are immunostimulatory in healthy human volunteers (Joseph et al., 1999) as well as in normal and tumor-bearing animals (Praveenkumar, et al., 1994b, 1999a, 1999b). Bone marrow cells from *Rasayana*-treated animals were found to proliferate in culture and the number of esterase-positive cells was high compared to untreated controls (Praveenkumar et al., 1999b). They are potent antioxidants both *in vitro* (Rekha et al., 1998) and in mice (Kumar et al., 1996) and healthy human volunteers (Joseph et al., 1999). *Rasayanas* were not directly cytotoxic to tumor cells but inhibited the growth of ascites and solid tumor in mice (Praveenkumar et al., 1994b). They inhibited methylcholanthrene-induced sarcoma in mice (Menon et al., 1996), and metastasis induced by B6F10 melanoma cell lines (Menon et al., 1997). We have previously reported that *Rasayanas* could increase survival, reduce leukopenia in the peripheral blood, and enhance bone marrow cellularity in mice treated with cyclophosphamide (Praveenkumar et al., 1994a). Similarly, we have shown that *Rasayanas* could augment the natural killer (NK) cell activity and antibody-dependent cellular cytotoxicity (ADCC), increase the number of bone marrow cells, and reduce leukopenia induced by radiation (Kumar et al., 1996). It has also been demonstrated that *Rasayanas* reduced myelosuppression in patients receiving chemotherapy and/radiotherapy (Joseph et al., 1999). In the present study, we provide further evidence to demonstrate the chemoprotective and radioprotective effects of *Rasayanas* against cyclophosphamide (CP) and radiation in mice.

MATERIALS AND METHODS

Brahma Rasayana (BR), *Narasimha Rasayana* (NR), *Amruthaprasham* (AP) and *Ashwaganda Rasayana* or *Ashwagandhadi Lehya* (AR) were purchased locally from Ashtavaidyan Thaikkattu Moos's Vaidyaratnam Oushadasala, Ollur, Trichur. The herbs used for the preparations were authenticated and standardization of each lot was done at the herbal pharmacy of Ashtavaidyan Thaikkattu Moos's Vaidyaratnam Oushadasala. The main ingredients of BR are *Embelica officinalis* (20%); *Terminalia chabula* (6.6%); *Urarira pitca, Desmodium gangeticum, Gmelina arborea, Solanum nigrum, Tribulus terrestris, Aegle marmelos, Premna tomentosa, Stereospermum suaveolens, Oroxylon indicum, Sida rhombilfolia, Boerhaavia diffusa, Ricinus communis, Vigna vexilata, Phaseolus adenanthus, Asperagus recemosus, Holostemma annulare, Leptadenia reticulata, Desmostachya bipinnata, Saccharum officinalum,* and *Oryza malampuzhensis* (0.4% each); and *Cinnamomum iners, Elettaria cardamomum, Cyperus rotundus, Curcuma longa, Piper longum, Aquilaria agallocha, Santalum album, Centella asciatica, Mesua ferrea, Clitoria ternate, Acorus calamus, Scirpus crossus, Glycyrrhiza glabra,* and *Embelia ribes* (0.2% each).

The main ingredients of AR are *Withania sominifera, Purerira tuberosa, Hemidesmus indicus, Ciminum cuminum,* and *Aloe barbidensis* (4.2%

each); *Vitis vinifera* (3.5%); and *Elettaria cardamomum, Zingiber officinale, Piper nigrum,* and *Piper longum* (0.7% each).

NR is prepared from the following herbs: *Acacia catechu, Plumbago xylanica, Xylia dolabriformis, Pterocarpus marsupium, Embelia ribes,* and *Semicarpus anacardium* (0.3% each); *Eclipta alba* (22%); and *Terminalia chebula, Embelica officinalis,* and *Terminalia belerica* (2.6%).

AP contains *Holstemma annulare, Vigna vexilata, Phaseolus adenanthus Glycyrrhiza glabra, Zingiber officinale, Asparagus recemosus, Boerhaavia diffusa, Sida retusa, Clerodendrum serratum, Macuna pruriens, Hedychium spicatium, Phylanthus neruri, Piper longum,* and *Vitis viniferra* (1.5% each); *Embelica officinalis* and *Purerira tuberosa* (23.8% each); *Saccharum officinalum* (7.1%); and *Piper nigrum, Cinnamomum zeylanica, Elettaria cardamomum, Garcinia morella,* and *Mesua ferrea* (0.14% each).

Cyclophosphamide was purchased from Khandelwal Industries, Bombay, India.

Inbred Swiss Albino mice were used for all experiments (6–8 weeks old; 20–25 g). They were housed in ventilated cages and fed with pelleted mouse chow (Lipton, Bombay, India) and water *ad libitum*.

CHEMOPROTECTIVE STUDIES

Animals were divided into six groups (7 per group). All groups received CP except the normal group of animals, which was left untreated. Of five CP-treated groups, four groups received *Rasayanas* and the other group was treated with only oral saline. CP was administered long-term (50 mg/kg body weight daily for 2 weeks) intraperitoneally to all five groups of animals. This dose has been demonstrated to cause bone marrow suppression and subsequent leukopenia in the peripheral blood, infection, wasting disease, and death of animals (Kuttan and Kuttan, 1993; Unnikrishnan et al., 1990). Simultaneously, each *Rasayana* was given orally (50 mg per amimal per day) to respective groups of animals daily for 2 weeks. This dosage was shown to be immunomodulatory (Praveenkumar et al., 1994b, 1999a, 1999b), and myeloprotective in mice treated with radiation (Praveenkumar et al., 1996) and chemotherapy (Praveenkumar et al., 1994a) and nontoxic (Praveenkumar, et al., 1994b). Animals treated with CP alone received saline orally. All animals were weighed before the start of the treatment and at various intervals until the end of the experiment for 30 days. The blood and liver were collected 48 hours after the last CP/*Rasayana* treatment. Serum and 10% liver homogenates were prepared and assayed for serum and tissue lipid peroxides (LP), alkaline phosphatase (ALP), and glutamate pyruvate transaminase (GPT).

RADIOPROTECTIVE STUDIES

The animals were divided into 6 groups of 10 each. One group received single whole-body irradiation only (400 rads per animal) and another group was normal animals without *Rasayana* or radiation treatment and served as control. The other groups received single doses of radiation and *Rasayanas* orally, respectively. Each *Rasayana* was administered orally 5 days prior to radiation (50 mg per animal per day) and continued for 1 month. The control animals received only saline. The animals were sacrificed at 48 hours to determine the early effect of *Rasayana* on radiation-induced damage or on day 7 to investigate the effect of *Rasayana* treatment on the recovery process after irradiation. Blood and tissues were collected and serum and 10% liver homogenates were used to determine lipid peroxides and serum GPT. The body weights of these animals were recorded on different days throughout the experiment.

DETERMINATION OF LIPID PEROXIDE

The tissue lipid peroxide (LP) was determined by the method of Ohkawa et al., (1979). To 200 μl of 10% (w/v) tissue homogenate, 200 μL of 8.1% sodium dodecyl sulfate (SDS), 1.5mL 20% acetic acid (pH 3.5), and 1.5mL of 0.8% thiobarbituric acid (TBA) solution were added. The mixture was combined with 4.0 mL of distilled water and then heated in an oil bath at 95°C for 1 hour. The tubes were cooled and 1.0 mL of distilled water and 5.0 mL of the mixture of n-butanol and pyridine (15:1; v/v) were added and vortexed vigorously. After centrifugation at 4000 rpm for 10 minutes, the organic layer was removed and the absorbance was measured at 532 nm. The values are obtained from the standard graph generated from 1, 1, 3, 3, tetraethoxy propane and expressed as nmoles of malondialdehyde (MDA) per milligram of protein. Protein concentration was determined by Lowry's method (Lowry et al., 1951).

Serum lipid peroxide was measured by the method of Yagi (1998). Serum (0.5mL) was mixed with 4 mL N/12 sulfuric acid and 0.5 mL of 10% phosphotungstic acid (PTA) and centrifuged. The supernatant was discarded and the same procedure was repeated with 2 mL PTA and 0.3mL of H_2SO_4. The sediment was suspended in 4 mL of distilled water and kept in an oil bath at 100°C with 1mL of TBA reagent (equal volumes of 0.67% TBA in distilled water and glacial acetic acid) for 60 minutes. The mixture was cooled, shaken vigorously with 5 mL of n-butanol:pyridine mixture (15:1; v/v), and centrifuged at 4000 rpm for 15 minutes. The organic layer was used to measure optical density (OD) at 532 nm. 1, 1, 3, 3, tetraethoxy propane was used as the standard to generate the standard curve. The curve was linear up to 20 nm/mL MDA.

DETERMINATION OF LIVER ENZYMES

ALP (King and Amstrong, 1980) and GPT (Bergmeyer and Berrnt, 1980) were determined by standard procedures. ALP was determined by estimating the amount of phenol liberated by the hydrolysis of disodium phenyl phosphate (DPP). It is colorimetrically measured using 4-aminoantipyrine, which turns purple in the presence of alkaline oxidizing agents. The absorbance was measured at 520 nm. A standard curve was prepared from various concentrations of phenol. GPT, however, was estimated by the amount of pyruvate formed during the transamination reaction. Pyruvate reacts with dinitrophenyl hydrazine (DNPH) forming its hydrazone. The absorbance was measured at 520 nm. The standard curve was generated from various concentrations of pyruvate.

STATISTICAL ANALYSIS

The change in body weight of animals treated with CP or radiation alone and that of *Rasayana*-treated groups on various days were compared by analysis of variance (ANOVA) repeated sample measurements. Other parameters were statistically compared between the mean values of normal, CP/radiation alone and the *Rasayana* treated groups by Student's *t* test/ANOVA. SPSS software (Chicago, IL) was used to analyze the data.

RESULTS

Chemoprotective effects of Rasayanas

The effect of *Rasayanas* on CP-induced weight loss in mice is shown in Table 1. There was a significant reduction in the body weight as a result of CP treatment compared to the no-treatment group ($p < 0.005$). By the completion of CP treatment, on day 16, there was a loss of 7.3 g from initial body weight for these animals. However, *Rasayana* administration significantly reduced this loss of body weight in animals treated with CP. In BR- and AR-treated animals the loss of body weight was only 3.6 g and 3.1 g, respectively, and was highly significant ($p < 0.005$). In other *Rasayana*-treated groups, such as NR and AP, there was a statistically significant reduction in loss of body weight ($p < 0.05$) in the initial stages of the treatment period, especially on day 9 and 12 respectively, as seen from the *post hoc* analysis. However, it was less significant in the later stages of the treatment, because the survival of these animals were poor. Controls, NR- and AP-treated animals did not survive more than 20 days (data not shown).

Table 2 shows the effect of *Rasayanas* on the wet organ weights of animals treated with CP. The animals treated with CP alone showed a significant reduction in liver (29%), kidney (39%), and spleen (64%) organ weights when compared to the no-treatment group. The administration of *Rasayanas* significantly reduced the loss of organ weight caused by CP treatment. In BR-treated animals, the weights of liver, kidney, and spleen were 39% ($p < 0.05$), 32% ($p < 0.001$), and 25% ($p < 0.005$) higher

TABLE 1. EFFECT OF *RASAYANAS* ON THE BODY WEIGHT OF ANIMALS TREATED WITH *CP*

	Body weight on various days (g)					
Treatment group	*0*	*3*	*6*	*9*	*12*	*16*
No treatment	28.3 ± 1.2 (−)	28.3 ± 1.0 (0)	28.4 ± 1.2 (0.1)	28.5 ± 0.9 (0.2)	28.8 ± 1.1 (0.5)	28.9 ± 1.0 (0.6)
CP alone	27.5 ± 0.3 (−)	27.3 ± 1.1 (−0.2)	24.9 ± 1.1 (−2.6)	24.7 ± 1.1 (−2.8)	22.2 ± 1.0 (−5.3)	20.2 ± 1.2 (−7.3)
BR + CP	30.4 ± 0.4 (−)	30.5 ± 0.5 (0.1)	28.10 ± 1.1 (−2.3)	27.9 ± 1.1 (−2.5)	26.7 ± 1.1 (−3.7)	26.8 ± 1.1* (−3.6)
NR + CP	28.6 ± 1.5 (−)	27.5 ± 1.3 (−1)	25.6 ± 1.6 (−2.9)	25.6 ± 1.6 (−2.9)	25.4 ± 1.5 (−3.1)	20.2 ± 1.2 (−8.3)
AR + CP	29.4 ± 1.3 (−)	28.4 ± 0.8 (−0.9)	28.5 ± 0.9 (−0.9)	27.8 ± 0.9 (−1.6)	26.4 ± 1.1 (−3.0)	26.3 ± 1.0* (−3.1)
AP + CP	27.4 ± 1.0 (−)	25.5 ± 1.6 (−2.0)	25.6 ± 1.7 (−1.9)	24.3 ± 1.7 (−3.2)	23.7 ± 1.7 (−3.8)	20.2 ± 1.2 (−7.3)

CP (50 mg/kg body weight) was administered intraperitoneally daily for 14 days. Simultaneously each *Rasayana* [50 mg/animal] was administered orally to respective groups of animals for 14 days. Each animal was weighed on different days both before and until 48 hours after the last dose of CP/*Rasayana* (see Materials and Methods in text). The values are mean ± SD of seven animals per group. Change in body weight at various time points were compared by ANOVA repeated sample measurements and *post hoc* analysis. The values in the parenthesis represent change in body weight from the initial value or day 0.

*$p < 0.005$.

CP, cyclophosphamide; BR, *Brahma Rasayana*; NR, *Narasimha Rasayana*; AR, *Ashwagandha Rasayana*; AP, *Amruthaprasham*; ANCOVA, analysis of covariance; SD, standard deviation.

than the CP alone treated group on day 16. Similarly, in the AR-treated group, 14%, 6%, and 41% ($p < 0.001$) increases were observed in liver, kidney, and spleen, respectively. In the NR-treated group, there was an increase of 24%, 30% ($p < 0.005$), and 100% ($p < 0.001$), and in AP-treated mice the increase was 17%, 5%, and 24% ($p < 0.005$) in liver, kidney, and spleen, respectively on day 16 compared to the no-treatment group of animals.

In mice, *Rasayana* treatment significantly reduced the levels of both tissue and serum LP induced by CP treatment (Table 3). When animals were treated with multiple doses of CP, the LP levels were elevated to 4-fold in the serum and 3-fold in the liver compared to the no-treatment group. Among the four *Rasayanas* studied, both BR and AP significantly reduced LP formation both in serum and liver ($p < 0.001$). There was a reduction by 55%, 25%, and

TABLE 2. EFFECT OF *RASAYANAS* ON THE ORGAN WEIGHTS OF MICE TREATED WITH *CP*

		Weight of organs (mg)		
Treatment group	*Body weight*	*Liver*	*Kidney*	*Spleen*
No treatment	18.6 ± 1.7	1019 ± 94.2	407 ± 97	57.9 ± 13
CP alone	22.4 ± 1.15	869 ± 100*	300 ± 64*	25.1 ± 2.5*
BR + CP	21.2 ± 0.68	1140 ± 51.6**	374 ± 68**	29.7 ± 3.6****
NR + CP	22.3 ± 0.68	1070 ± 106****	387 ± 108	50.0 ± 1.6**
AR + CP	19.0 ± 0.89	840 ± 131	270 ± 106	30.0 ± 2.5**
AP + CP	20.2 ± 0.68	917 ± 104****	284 ± 43	28.0 ± 2.5****

CP (50 mg/kg body weight) was administered intraperitoneally daily for 14 days. Simultaneously each *Rasayana* [50 mg per animal] was administered orally to respective groups of animals for 14 days. Animals of each group were sacrificed 48 hours after the last dose of CP/*Rasayana* and the wet weight of each organ was measured (see Materials and Methods). The values are mean ± SD of 7 animals per group. Statistical analysis was done by comparing the mean organ weight per gram body weight of the animal between the CP alone treated and *Rasayana*-treated animals.

*$p < 0.001$ compared to the no-treatment group; **$p < 0.001$; ***$p < 0.05$; ****$p < 0.005$.

CP, cyclophosphamide; BR, *Brahma Rasayana*; NR, *Narasimha Rasayana*; AR, *Ashwagandha Rasayana*; AP, *Amruthaprasham*; SD, standard deviation.

TABLE 3. EFFECT OF *RASAYANAS* ON THE BIOCHEMICAL PARAMETERS OF TISSUE INJURY INDUCED BY *CP*

	Biochemical parameters					
	Glutamate pyruvate transaminase		*Alkaline phosphatase*		*Lipid peroxide*	
Treatment	*Serum (U/mL)*	*Liver (U/mg protein)*	*Serum (KAU/100 mL)*	*Liver (KAU/mg protein)*	*Serum (nmols/mL)*	*Liver (nmols/mg/protein)*
No treatment	217 ± 5.2	91 ± 12	15.2 ± 1.2	5.6 ± 2	1.3 ± 0.2	1.6 ± 0.2
CP alone	1123 ± 26*	239 ± 31*	28 ± 0.5	18 ± 9.0*	5.24 ± 0.70*	5.48 ± 0.94*
BR + CP	341 ± 3.1**	144 ± 22**	21 ± 0.9**	7.4 ± 0.7**	2.37 ± 0.59**	2.53 ± 0.49**
NR + CP	1118 ± 11	211 ± 20	27 ± 1.3	15.9 ± 4.4	4.90 ± 0.58	5.04 ± 0.79
AR + CP	556 ± 3.8**	219 ± 92	23 ± 0.9**	10.9 ± 2.7***	3.92 ± 0.68	5.94 ± 0.88
AP + CP	1121 ± 31	263 ± 18	28 ± 0.5	19.8 ± 4.5	2.08 ± 0.36**	1.78 ± 0.52**

CP (50 mg/kg body weight) was administered intraperitoneally daily for 14 days. Simultaneously each Rasayana (50 mg/animal) was administered to respective groups of animals for 14 days. Animals of each group were sacrificed 48 hours after the last dose of CP/*Rasayana* treatment and liver and serum was collected (see Materials and Methods). The values are mean ± SD of 7 animals per group and compared between CP alone treated and *Rasayana*-treated groups.

*$p < 0.001$ compared to the no-treatment group; **$p < 0.001$; ***$p < 0.005$.

CP, cyclophosphamide; BR, *Brahma Rasayana*; NR, *Narasimha Rasayana*; AR, *Ashwagandha Rasayana*; AP, *Amruthaprasham*; SD, standard deviation.

60% in the formation of serum LP in the BR-, AR-, and AP-treated groups, respectively, by day 16 compared to CP-treated controls. However, only BR and AP were shown to inhibit liver LP formation in CP-treated animals. There was a reduction of lipid peroxide by 54% and 68% ($p < 0.001$) in BR- and AP-treated groups, respectively. More interestingly, the levels of LP in the AP-treated group are comparable to that of the levels in the no-treatment group even after long-term CP treatment.

Treatment of animals with *Rasayanas* significantly reduced the serum and tissue levels of enzymes that are good markers of liver damage (Table 3). Animals treated with CP alone demonstrated significant damage to the tissues as seen from the elevated levels of serum and liver GPT and ALP. There was a significant elevation of serum and liver GPT levels in this group by 6- and 3-fold, respectively, compared to the no-treatment group ($p < 0.001$). Among the four *Rasayanas* studied, only BR and AR were found to reduce the level of serum GPT (70% and 51%, respectively) significantly ($p <$ 0.001) compared to animals treated with CP alone. However, only BR could prevent liver damage significantly ($p < 0.001$] as seen from reduced levels of liver GPT (40%).

Treatment with CP alone produced increased levels of serum and liver ALP compared to the no-treatment group. There was an elevation of 2- and 3-fold in the serum and liver, respectively, in this group. However, treatment with *Rasayanas* such as BR and AR significantly reduced the levels of ALP in the serum (25%, [$p < 0.005$] and 18% [$p < 0.05$], respectively and liver ALP (59%, [$p < 0.001$] and 39%, [$p < 0.001$]).

Radioprotective effects of Rasayanas

The effect of *Rasayanas* on the body weight of animals treated with radiation is given in Table 4. There was a significant reduction in the body weight of control animals treated with radiation alone ($p < 0.001$). There was reduction of 3.4 g by day 29 in these animals. However, in the BR- and AR-treated groups, there was only a reduction of 0.7 and 1.1 g by day 29, which was highly significant ($p < 0.001$). In the NR- and AP-treated groups there was only a loss of weight of 2.5 g and 2.3 g, respectively. However this reduction was not statistically significant.

Rasayanas significantly reduced the serum LP formation in mice treated with radiation (Table 5). In animals treated with radiation alone, there was a significant increase in serum LP after 2 days of radiotherapy (6-fold increase) and reduced to 4-fold by day 7. *Rasayanas*, especially, AP, inhibited serum LP formation almost completely in irradiated animals when

TABLE 4. EFFECT OF *RASAYANAS* ON BODY WEIGHT OF ANIMALS WITH RADIATION

	Body weight of animals on various days (gm)					
Treatment	*0*	*6*	*14*	*19*	*24*	*29*
No treatment	22.3 ± 1.5 (−)	22.5 ± 1.3 (0.2)	22.9 ± 1.5 (0.6)	23.1 ± 1.4 (0.8)	23.6 ± 1.5 (1.3)	23.8 ± 1.3 (1.5)
RT alone	23.7 ± 2.4 (−)	22.1 ± 3.6 (1.6)	22.2 ± 3.4 (1.5)	21.6 ± 4.2 (2.1)	21.6 ± 3.6 (2.1)	20.3 ± 3.8 (3.4)
RT + BR	23.9 ± 3.5 (−)	22.5 ± 3.3 (1.4)	23.6 ± 2.9 (−0.3)	23.2 ± 2.9 (−0.7)	23.6 ± 1.1 (−0.3)	23.2 ± 1.5* (−0.7)
RT + NR	23.4 ± 2.2 (−)	22.8 ± 3.1 (−0.6)	22.1 ± 3.9 (−1.3)	21.6 ± 1.9 (−1.8)	21.5 ± 3.1 (−1.9)	20.9 ± 3.8 (−2.5)
RT + AR	22.7 ± 2.1 (−)	21.3 ± 3.6 (−1.4)	21.9 ± 2.1 (−0.8)	21.8 ± 2.3 (0.9)	21.2 ± 2.3 (−1.5)	21.6 ± 2.3* (−1.1)
RT + AP	23.2 ± 3.7 (−)	21.7 ± 4.2 (−1.5)	21.3 ± 6.2 (1.9)	21.2 ± 6.0 (−2.0)	21.3 ± 6.2 (−1.9)	20.9 ± 3.1 (−2.3)

All groups of animals were exposed to a single whole-body γ irradiation except the untreated controls. Each *Rasayana* (50 mg/animal) was administered to respective groups of animals 5 days before RT and continued until the end of the experiment. The weight of each animal was measured both before the start of the experiment and on different days for 1 month. The values are mean ± SD of 10 animals per group. Change in body weight at various time points were compared by ANOVA repeated sample measurements and *post hoc* analysis. The values in the parentheses represent change in body weight from the initial value or day 0.

*$p < 0.005$.

RT, radiotherapy; BR, *Brahma Rasayana*; NR, *Narasimha Rasayana*; AR, *Ashwagandha Rasayana*; AP, *Amruthaprasham*; SD, standard deviation; ANOVA, analysis of variance.

compared to the RT alone group. AP inhibited serum LP by 56% 2 days after irradiation, and on day 7 there was an inhibition of 40% compared to irradiated controls ($p < 0.001$). Other *Rasayanas* such as BR, NR, and AR significantly inhibited the lipid peroxidation on day 7 by 33%, 71%, and 35%, respectively, which is highly significant. However, on day 2 BR, NR, and AR had no significant effect on reducing the LPs. However serum LP levels were significantly low when compared between day 2 and 7 in *Rasayana*-treated groups.

Inhibition of serum LPs by *Rasayanas* was further supported by significant reduction in the formation of tissue LPs. Among the four *Rasayanas* studied, AP was found to be the most potent in reducing the LP formation in the liver. AP administration significantly reduced the

TABLE 5. EFFECT OF *RASAYANAS* ON THE BIOCHEMICAL PARAMETERS OF TISSUE INJURY INDUCED BY RADIATION

	Serum GPT (U/mL)		*Serum lipid peroxide [nmol/mL]*		*Liver Lipid peroxide [nmol/mg protein]*	
Treatment	*Day 2*	*Day 7*	*Day 2*	*Day 7*	*Day 2*	*Day 7*
No treatment	197 ± 16	199 ± 27	0.95 ± 0.02	0.93 ± 0.1	0.72 ± 0.07	0.75 ± 0.06
RT alone	569 ± 21*	545 ± 82*	5.50 ± 1.00*	3.79 ± 1.10*	2.00 ± 0.26*	1.39 ± 0.09*
RT + BR	313 ± 12**	202 ± 29**	5.86 ± 2.31	2.55 ± 1.09**	2.76 ± 0.9	0.79 ± 0.06**
RT + NR	377 ± 69**	233 ± 43**	4.65 ± 2.01***	1.10 ± 0.32**	1.67 ± 0.77	0.40 ± 0.013**
RT + AR	421 ± 39****	221 ± 42**	4.67 ± 0.33***	2.40 ± 0.69**	1.73 ± 0.21	0.87 ± 0.28**
RT + AP	367 ± 55**	249 ± 25**	2.38 ± 1.53**	2.28 ± 0.59**	0.87 ± 0.61**	0.84 ± 0.23**

All the groups of animals were exposed to a single whole body gamma irradiation except the untreated controls. Each Rasayana [50 mg/animal] was administered to respective groups of animals 5 days before RT and continued till the end of the experiment. The values are mean ± SD of seven animals per group. The animals were sacrificed on day 2 and day 7 after RT and serum and liver homogenates were assayed for GPT and lipid peroxides. The values are the mean ± SD of 5 animals per group and compared between RT alone treated and Rasayana treated groups.

*$p < 0.001$ compared to the no-treatment group; **$p < 0.001$; ***$p < 0.05$; ****$p < 0.01$.

RT, radiotherapy; BR, *Brahma Rasayana*; NR, *Narasimha Rasayana*; AR, *Ashwagandha Rasayana*; AP, *Amruthaprasham*; GPT, glutamate pyruvate transaminase.

lipid peroxides on both day 2 and 7 by 57% and 40% when compared to radiation-treated controls ($p < 0.001$) and is comparable to the LP levels in normal healthy animals (0.75 nmol of MDA per milligram of protein). However, other *Rasayanas* such as BR, AR, and NR effectively and significantly reduced LPs in the liver only on day 7 by 43%, 71%, and 40% ($p < 0.001$).

When the animals were γ-irradiated the serum GPT levels rose 3-fold on day 2 and the level was maintained at 3-fold even on day 7. Serum GPT levels are also reduced in irradiated animals demonstrating the tissue protection conferred by *Rasayanas* (Table 5). All *Rasayanas* significantly reduced the serum GPT levels 2 days after irradiation. Serum GPT levels were reduced by 48% ($p < 0.001$), 37% ($p < 0.001$), 29% ($p < 0.005$), and 38% ($p < 0.001$) in BR, NR, AR, and AP treated animals, respectively. By day 7, serum GPT levels in all *Rasayanas*-treated groups reached a level equal to or less than that of normal healthy animals, which is highly significant ($p < 0.001$).

DISCUSSION

The Indian indigenous system of medicine, Ayurveda, has identified several herbs as well as complex herbal drug preparations as useful in the treatment, prevention, and for improvement of the quality of life of both healthy and diseased individuals. Many of them are well known for their antioxidant and immunomodulatory properties (Scartezzini and Speroni, 2000). *Rasayana* (for review, see Vayalil et al., 2002) is a branch of Ayurveda, literally means "the path of juice" or "juice-incorporate," (Madhihassan, 1981) the goal of which is to nourish, restore, and balance the body functions. According to Ayurvedic scripts, *Rasayana* is generally used to rejuvenate general health of the body (Bhishagratna and Kaviraj Kunjalal, 1991; Kaviratna and Sharma, 1996, Srikantha Murthy, 1991–1992) or attempts to achieve the body's maximum potential. The restoration of balance or potential within the human body by maintaining or replenishing the "juiciness" is called *Rasayana* (Mahdihassan, 1981, 1991). One of the ways by which rejuvenation may be attained is through medicines. *Rasayana* medicines (medicinal rejuvenation) utilize a group of individual herbs or medicinal drug preparations that are customarily complex mixtures based mostly on herbal products (Bhishagratna and Kaviraj Kunjalal, 1991; Kaviratna and Sharma, 1996, Srikantha Murthy, 1992). According to classic Ayurvedic texts, *Rasayana* therapy arrests aging, enhance intelligence, memory, strength, youth, luster, sweetness of voice, and vigor. They are supposed to nourish blood, lymph, flesh, adipose tissue, and semen and thus prevent degenerative changes and illness (Bhishagratna and Kaviraj Kunjalal, 1991; Kaviratna and Sharma, 1996, Srikantha Murthy, 1992). *Rasayana* is thought to improve the metabolic processes, which results in best possible biotransformation and produce best-quality bodily tissues and eradicate senility and other diseases of old age (Singh, 1990). Most of these *Rasayanas* can be used regularly as a food for maintaining balanced mental and physical health. *Rasayanas* may be used either alone or along with other modalities of treatment as an adjuvant (Singh, 1990). We have already reported that *Rasayanas* are potent myeloprotective agents against chemotherapeutic agents (Praveenkumar et al., 1994) and radiation (Kumar et al., 1996; Rekha et al., 2000) in mice. *Rasayanas* are also effective myeloprotectors in humans undergoing chemotherapy and or radiotherapy (Joseph et al., 1999). In the present study, we provide additional evidence to support the chemoprotective and radioprotective activity of four *Rasayanas* widely used in Indian indigenous system of medicine.

The nontarget effects of chemotherapy and radiotherapy instigated the development of methods to improve the therapeutic efficiency of these treatment modalities. One of the strategies is to increase the resistance of the normal cells to effects of cytocidal therapy. In the present study, we have shown that *Rasayanas* could protect the normal tissues as seen by increased body weight and organ weights. Moreover, it protected the normal tissues from injuries induced by cytotoxic therapies as seen from the decreased levels of liver enzymes, which are good markers of tissue damage.

One of the mechanisms by which chemotherapeutic drugs and radiation act on and kill malignant cells is by free radicals. However, these free radicals are not target-specific. Free radicals generated in one area can leak into the circulation and can act on other cell components and can cause inimical reactions and toxicities in the host. Therefore, in order to ameliorate the toxicities caused by these treatment strategies, agents that can scavenge these free radicals or reactive metabolites without affecting the outcome of the treatment would be beneficial to the host as well as enhance the efficiency of the treatment. In this report, we have demonstrated the inhibition of LP formation by *Rasayanas* in both CP- and radiation-treated animals, which is highly significant compared to control animals. One of the most striking features among the four *Rasayanas* studied was the potent antioxidant property of AP throughout the experiment. No significant increase in LP either in the serum and liver was seen in this group. The values are comparable to that of normal healthy controls. AP is, however, not a good radioprotector or chemoprotector because of poor survival of the animals, increased loss of body weight, reduced organ weights, and tissue enzyme levels. This implicates that AP contains herbs or compounds having high antioxidant activity, which is currently elusive. Hence, it is worthwhile to analyze AP or herb(s) used for its preparation for the active principles that confer the antioxidant property. Thus, further studies are warranted. Moreover, the *Rasayanas* used in the present study are complex herbal formulations, which contain thousands of phytochemicals in various proportions. At present, we do not know the phytochemical(s) involved in these activities. Moreover, we do not know whether the separation of individual components will affect the efficiency of these drug preparations. In addition, the proportion of each component and the combination of herbs may also be important in the action of each drug preparation. For instance, in BR and AP, the main ingredient is *Emblica officinalis* (20% and 24%, respectively). Although both *Rasayanas* are good antioxidants, only BR was found effective as a radioprotective and chemoprotective agent. Therefore, the beneficial effect of these formulations may be because of a combined effect of all the components, in a certain proportion and combination that act synergistically in the system in favor of the host.

Currently, the mechanism by which *Rasayanas* confer radioprotective and chemoprotective effects in mice as well as in humans is still obscure. One of the mechanisms may be by scavenging free radicals. Bone marrow protection as well as its rapid regeneration after cytotoxic therapy may also be important in protecting the animals from the toxic treatment. Among the four *Rasayanas* used in the study, only BR and AR were found to induce bone marrow cell proliferation in animals. The studies have shown that only BR and AR were found effective as myeloprotective agents against chemotherapy and radiation. They also helped to recover rapidly from the immunosuppression induced by toxic treatments. This may partially explain the difference in protective effects among four *Rasayanas*. However, further studies are needed, such as the effect of CP/RT combined with *Rasayana* on tumor-bearing animals, so that its use in cancer therapy may be fully appreciated and may help to elucidate the mechanism of action of *Rasayanas*.

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